USER GUIDE / EDITION 12 OCTOBER 2026

EUROFIGHTER IFE FUEL CALCULATOR

Operating guide and capability overview
Version 1.9.23 | Simulator mission planning

Purpose

Plan a mission from initial inputs to a printable Data Card: route, timing, fuel, holds, AAR, diverts and fuel thresholds. Save the complete current plan as JSON and restore it on another device.

Capabilities at a glance

AREA	CAPABILITY	P.
MISSION / ROUTE	Mission details, SimBrief import, waypoint editor and vertical route profile.	02
FUEL / CRUISE	Forward calculation from entered start fuel; separate cruise calculator.	03
TIMING / RECOVERY	Hard times, holds, WP IN/OUT, Joker, Bingo and two diverts.	04
AAR	AUTO/MANUAL, gross onload, transfer rate, transfer time and event burn.	05
BACKUP / RESTORE	Complete JSON export/import and local mission storage.	06
OUTPUT / INTERFACE	Data Card, AAR ticket, tablet controls, language and appearance.	07
WORKFLOW / LIMITS	Planning sequence, model limitations, validation and terminology.	08
ONLINE / UPDATES	Open the web app, check versions and update safely.	09

Document scope: a practical product and operating overview. This is not an official flight document or military regulation. It describes implemented features in this software version, not planned extensions.

Illustrated examples: AAR setup p. 10; ticket p. 11; fuel thresholds p. 12; holds p. 13; glossary p. 14. AAR/fuel background pp. 15-19; sources p. 20.

Open the current app

ife-eufi-fuelcalc-test.topstoni81.workers.dev
Bookmark or install this web app to the home screen. ZIP/HTML downloads are separate copies. Update guidance: page 9.

02 / SECTION

Mission and route

Create, import and edit a plan while retaining control of its inputs.

Mission setup

Enter Mission Name, Mission Type, UTC date, Start Z, Aircraft, Callsign, Departure/Arrival, airfield elevations and mission notes. Start fuel, taxi time, reserve and a fuel consumption surcharge are editable. Airfield elevations can be taken from the local database or entered manually.

ENTRY POINT	BEHAVIOUR
NEW MANUAL MISSION	Start a manual plan with editable intermediate waypoints.
LOAD DEMO	Load the existing demonstration route with DIST/TIME legs and two diverts. The app still starts with the demo.
SIMBRIEF	Import the latest OFP using a Pilot ID or alias. Imported data include date, route, coordinates, altitudes, Mach and wind. Planning values remain editable afterwards.
LOAD PRESET	Explicitly load a separately offered mission preset. Selecting a preset in the list alone does not change the active mission.

Route / leg editor

Fields are ordered **WP - ALT ft - Mach - time/ETA Z - NM - Hold/WP OUT**, followed by wind, mode, manual leg minutes, ETE, leg fuel, recovery requirement, FOB and actions. Coordinates remain part of the internal route state and backup.

Intermediate waypoints can be added, deleted or moved up/down. AAR, Bingo and Joker references follow their associated waypoint when it moves. Deliberately reordering a route may recalculate affected DIST legs from available coordinates.

Distance or time: choose the governing input

DIST: NM is specified; ETE follows from speed, wind and altitude changes. **TIME:** entered leg minutes govern the calculation; leg distance is derived from them. Editing NM or MIN selects the corresponding mode. Holds, AAR duration and hard times do not change leg NM or coordinates.

Precision and editing

Mach is displayed with no more than two decimal places. Full precision is retained internally. Route edits commit on Enter or when leaving the field. Leaving a rounded field unchanged does not overwrite its precise internal value.

03 / SECTION

Fuel and performance

Separate the planned fuel state from the required minimum.

Start / On Board Fuel is an input. The app calculates fuel states forwards from this value. A desired landing quantity or AAR target does not back-calculate and overwrite start fuel.

DISPLAY	MEANING
FOB START / ON BOARD	Entered fuel on board before the planned ground consumption.
BLOCK REQ	Planning requirement including route, holds, reserve, divert/recovery and the net AAR effect.
T/O REQ	Required take-off fuel; this is not the actual or planned tank content.
PLANNED T/O FOB	Planned fuel on board after subtracting ground consumption from start fuel.
UPLIFT / SPARE	Additional fuel needed to meet the block requirement, or the difference between on-board fuel and the planning requirement.
FOB IN / OUT / LANDING	Calculated fuel states at waypoints, after events and at landing.

Calculation basis

Climb/descent and cruise portions are calculated separately for each leg. Mach is converted to TAS using ISA; the entered wind component affects groundspeed. The fuel curve interpolates between measured points and uses the stored estimation model outside the measured Mach range.

The supplied EF2000 profile contains tank capacity and ground, climb and descent rates. The active Mach range is 0.48 to 0.95; model altitude extends to 55,000 ft. The airborne consumption surcharge applies to flight phases including holds and AAR burn. Ground consumption is calculated separately from taxi minutes and the ground rate.

Separate cruise calculator

Calculate by total distance or total flight time. Distance may come from the current route, the direct DEP-ARR distance or manual entry. Set cruise altitude, Mach, wind and both airfield elevations. Outputs include climb, cruise, descent and total fuel, fuel flow, ETE, distance and block requirement including reserve and recovery/divert.

Legs too short for an altitude change, non-positive groundspeed and out-of-model values are flagged for review. Missing performance data are not filled by inventing a new aircraft profile.

04 / SECTION

Timing, holds and recovery

Time constraints and position-dependent fuel thresholds.

Hard times and holds

Start time and ETA use Zulu/UTC. A manual waypoint ETA acts as a hard-time anchor. Additional waiting time is treated as a hold with fuel consumption, preferably at a planned hold waypoint before the anchor. Unreachable times are flagged as conflicts; distances are not invented to make them fit.

A hold changes **time and fuel**. Each affected waypoint has separate WP IN time/FOB, hold duration/burn and WP OUT time/FOB. Consumption uses the hold waypoint altitude and Mach plus the airborne consumption surcharge. Subsequent ETAs shift accordingly.

Joker and Bingo

FUNCTION	LOGIC
JOKER @ WP IN	The hold at the selected waypoint is still included in the remaining requirement.
JOKER @ WP OUT	The hold at the selected waypoint is complete and is not counted again.
JOKER BUFFER	An editable additional buffer above the calculated requirement at the selected reference point.
BINGO REFERENCE	Position-dependent requirement for the remaining mission plus divert, recovery extra and landing reserve. At arrival, remaining mission fuel may be zero.
MANUAL / AUTO	Bingo and Joker can be calculated or replaced with explicit manual thresholds. The app shows the reference point, planned FOB and margin.
BINGO 2 / B2 RTB	A separate entered threshold with its own RTB airfield on the AAR plan.

Divert and reserve

Plan two alternate airfields. Search supports ICAO, name, country, favourites and recent airfields. Distance and elevation may use local data or manual entries. Divert altitude, Mach, wind, Missed Approach / Recovery Extra and Landing Reserve remain editable.

USED DIVERT can automatically select the higher valid fuel requirement or explicitly use Divert 1/2. A preset may contain labelled briefing fuel values. AAR onload that has not yet been received is not credited as fuel already available when determining conservative remaining fuel thresholds.

05 / SECTION

AAR: automatic planning

A330 MRTT, measurement-based transfer rates and separate time quantities.

Enable PLAN AAR and select the waypoint. Review FL, KIAS, configuration, event duration, ARIP/ARCP, callsign and frequency. **A330 MRTT is fixed.** Choose an AUTO target or manual gross onload. Normal use does not require selecting a measurement button or entering a rate.

FIELD	MEANING
RATE MODE: AUTO	Default. Matching measurement curves are selected automatically. Comparable points are interpolated within their coverage; reference and fuel-band estimates are labelled.
RECEIVER FUEL FLOW	Own consumption from AAR altitude, KIAS and the available consumption model. CLEAN explicitly uses the 3-BAGS model as an approximation.
RATE MODE: MANUAL	Explicit override of effective gross rate in kg/min. RATE SOURCE identifies the basis actually used.
TRANSFER / EVENT	TRANSFER TIME is fuel-transfer time. AAR EVENT DURATION also includes planned orbit/waiting time. The two remain separate.

Measurements and calculation

Six supplemental profiles with 30 checkpoints measure net fuel gain over time. Gross rate = measured net rate + modeled receiver consumption. The app integrates individual fuel-fill segments; the displayed effective average can differ from a complete Excel measurement-run average.

```
MANUAL: Gross Onload remains exactly the entered value

AUTO: G = max(0, Target OUT - IN + F × EventDuration)

AAR Burn = TransferBurn + OrbitBurn

FOB OUT = FOB IN + Gross Onload - AAR Burn
```

For a constant rate R, transfer time = G/R. Automatic measurement curves integrate each crossed fuel segment. TARGET FOB OUT applies at the end of the entire event. Zero onload may be correct.

Warnings and limits

Rate ≤ receiver flow, insufficient event duration and tank-capacity violations are flagged. Transfer is modeled first, then orbit; peak tank content is checked at transfer end. FOB AFTER LOAD is accounting only. Missing usable rate: N/A / RATE REQUIRED, without inventing a transfer time. Start fuel, leg NM and coordinates are not changed to fit the result.

06 / SECTION

Complete mission backup

Plan once, save to a file and restore on another device.

EXPORT CURRENT MISSION / IMPORT MISSION

Both controls are under MISSION. Export creates a JSON file of the current state; SAVE LOCAL is not required first. Import validates the file, restores the inputs and recalculates the plan. A successful import displays **Mission successfully imported**.

SAVED	COVERAGE
Mission inputs	Name/type, date, start, aircraft, callsign, DEP/ARR, elevations, notes and all calculation-relevant form values.
Internal route state	Order, names, available coordinates, ALT, exact Mach values, wind, NM, modes, manual times, hard times, planned times and holds.
Events and thresholds	AAR switch/reference and all AAR inputs including rate/source; Bingo/Joker switches, WP references, IN/OUT basis and manual values.
Recovery / performance	Diverts, reserve, manual distances/elevations, cruise settings, metadata and the aircraft profile, fuel curve and navigation data in use.
Results	A reference snapshot of leg, fuel, AAR and summary values. It never replaces recalculation on import.

Version handling

Current format: **schemaVersion 3**, appVersion 1.9.23, model **ife-forward-auto-3**. AUTO mode and measurement data are archived. Schema 1/2 migrations preserve explicit rates as MANUAL; former blank rates adopt AUTO with installed measurements. Unknown versions and missing required fields are rejected before loading.

File limit: 5 MB. The complete format supports 2 to 2,000 waypoints and does not truncate a backup route to the earlier 40-WP limit. Blank inputs stay blank; unknown coordinates are not estimated.

Local storage

SAVE LOCAL and LOAD LOCAL save/restore on the same device. New local saves use the same complete structure as JSON export; older browser saves remain readable through the existing loader. Export, file import and preset loading do not overwrite the saved local mission. Deliberate SAVE LOCAL replaces the local storage slot.

JSON is the authoritative complete backup format. PDF is for reading, not reimporting. An additional CSV/Excel export is not implemented in this version.

07 / SECTION

Output and interface

Cockpit-style information with display precision separate from calculation precision.

Data Card and AAR ticket

The Data Card combines the mission header, fuel order, Bingo/Joker, diverts/reserve and AAR. The flight log includes WP, ALT, Mach, wind, leg NM, ETE, WP IN, Hold/WP OUT, leg fuel, recovery requirement and FOB IN/OUT. An actual-fuel column (IST kg) is provided for recording observations.

PRINT / SAVE PDF uses the browser print dialogue. Print the complete Data Card or the separate AAR ticket. Use the JSON backup for full restoration.

Tablet controls and detail views

On touch devices or narrower screens, route rows become expandable waypoint cards. Fields start expanded. Tap the full waypoint name to open or close the fields. The desktop table remains available, with a waypoint column that stays visible during horizontal scrolling.

The vertical route profile shows the altitude sequence and marks AAR. Selecting a waypoint in the profile opens leg, timing, consumption and fuel-state details. Key fuel displays also provide detail views.

AREA	FUNCTION
LANGUAGE	English is the default. Settings → Language switches between English and German. Previously saved choices remain effective; times use UTC.
LAYOUT	OPS and OPERATIONAL EFB display variants.
THEMES	Night Cyan, Night White, Ops Green and Day High Contrast.
LOCAL OPERATION	Browser-based app with a service-worker cache. Calculations run locally; offline use requires previously loaded/available app files.
DATA STATUS	Profile/fuel data, AAR measurement status and AIRAC calendar cycle. The cycle does not certify that navigation data are current or approved.
PRIVACY	No app account or tracking SDKs. SimBrief is contacted on explicit import. Local app data can be deleted after confirmation.

Optional mission preset

RHINE SENTINELLE - GHOST - 10.10.2026 can be loaded separately. It contains the stored route, timing, holds, AAR and briefing fuel values. It remains optional and is not the app default. The general AUTO/MANUAL rate logic and labelled estimates also apply to this preset.

08 / SECTION

Workflow and limitations

Use the simulator model with its assumptions in view.

SEQUENCE	CHECK
01 / CREATE	Choose a manual mission, SimBrief or an optional preset. Check date, start, airfields and start fuel.
02 / ROUTE	Check order, coordinates, altitudes, Mach, wind and DIST/TIME modes.
03 / TIMING	Set hard times and holds. Check WP IN/OUT and timing conflicts.
04 / RECOVERY	Set diverts, Used Divert, Recovery Extra, reserve and Joker/Bingo references.
05 / AAR	Check AUTO/MANUAL, gross/target, configuration, rate and event duration. Resolve time and tank warnings.
06 / SAVE	Review recalculated results, export JSON and print the Data Card/AAR ticket to PDF if needed.

Model limitations

The calculator uses simulator measurements, interpolation and identified model assumptions. Below 3,000 ft it uses the 3,000-ft curve; a separate measured low-level dataset for every altitude is not available. Weather, weight and configuration are not covered by a complete real-world performance database.

Gross AAR rates are derived from measured net gain and modeled receiver burn. CLEAN uses the 3-BAGS consumption model as an approximation. 3 BAGS is currently measured at FL230 / 285 KIAS; other conditions use a labelled reference estimate. Interpolation requires comparable measurement coverage.

Validation and document basis

Checks cover AUTO/MANUAL, six complete measurement curves, interpolation, capacity, unchanged geometry, export-reset-import and SimBrief 793765. Focused update tests cover version comparison, offline failures and mission restoration. These tests are not real-world aircraft performance approval.

Basis: app package v1.9.23; app.js, logic-core.js, mission-backup.js, index.html, data files, README and related regression tests. This guide describes the implemented scope as of 12 October 2026.

Terminology

FOB Fuel on Board · **ETE** Estimated Time Enroute · **ETA** Estimated Time of Arrival · **AAR** Air-to-Air Refuelling · **WP** Waypoint · **NM** Nautical Miles · **RTB** Return to Base · **TOT** Time on Target

SIMULATION ONLY. Not an official flight document or approved for real-world operational planning.

09 / SECTION

Online app and updates

Recommended access: the official web app or its home-screen installation.

Open the app instead of downloading a ZIP

ife-eufi-fuelcalc-test.topstoni81.workers.dev

Bookmark this address or add it to your home screen through the browser. This PWA still uses the official address. A GitHub ZIP or local HTML copy is a separate, static copy.

What the app detects from v1.9.23

Opening the app checks the public GitHub version file. Returning to the app checks at most hourly; reconnecting also triggers a check. SETTINGS → APP UPDATES & USER GUIDE provides CHECK FOR UPDATES. No mission data or Pilot ID is sent with version checks.

SITUATION	ACTION
OFFICIAL APP / PWA	A newer version offers UPDATE NOW. The current plan is secured in this tab’s session storage, the app reloads and the plan is restored. SAVE LOCAL is untouched. No forced reload while planning.
SEPARATE COPY	A persistent notice points to the official app. Use EXPORT CURRENT MISSION, then OPEN LATEST APP and IMPORT MISSION. A browser cannot replace downloaded source files by itself.
OFFLINE / FAILURE	The app does not claim to be current. The available version remains usable where its files are cached. Choose CHECK FOR UPDATES after reconnecting.
OLDER THAN v1.9.23	These old copies do not have update detection. Switch once to the official address or obtain the new version.

Mission preservation and recovery

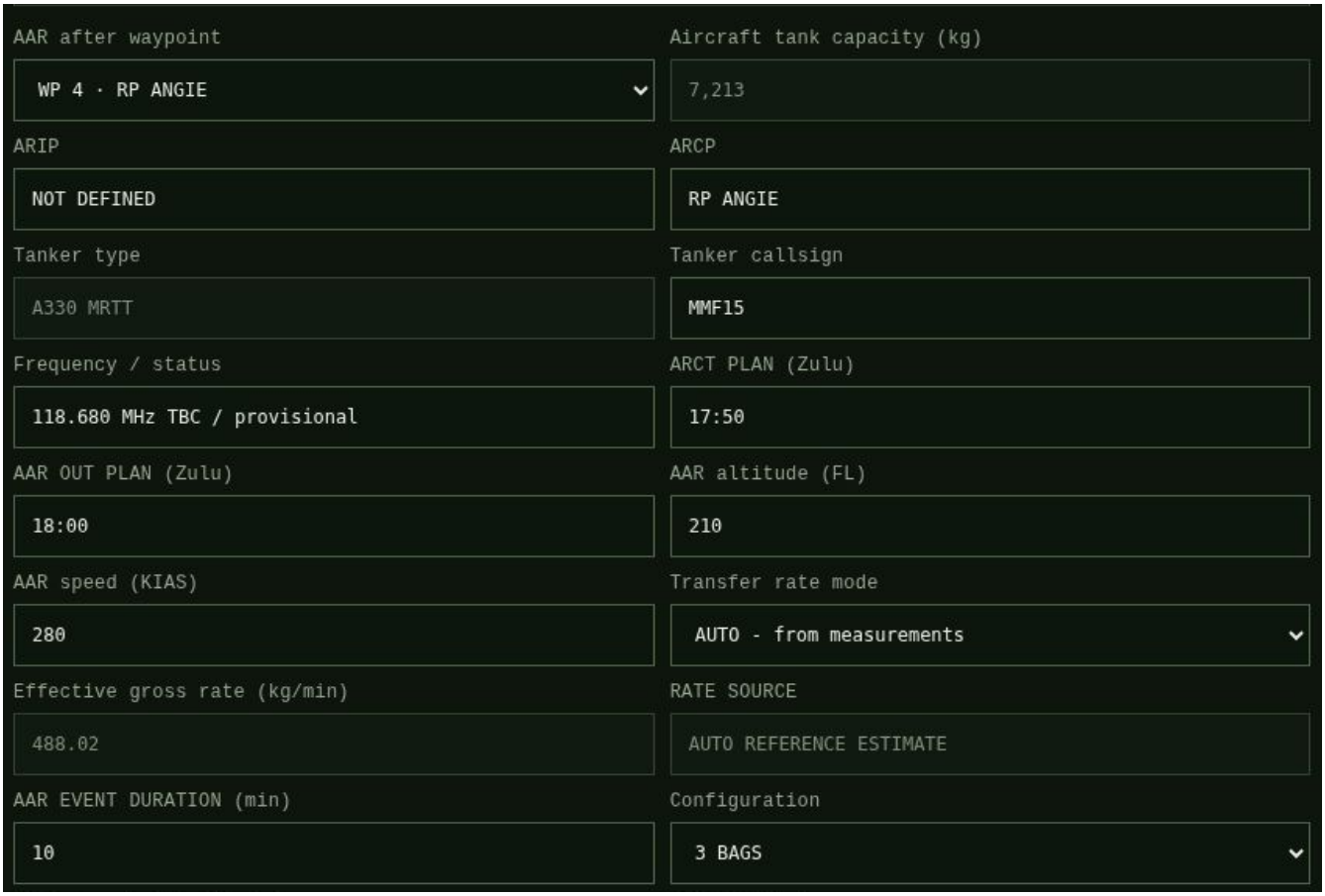
Restoration includes inputs and archived performance data. It does not replace an independent JSON backup. If automatic restoration fails, the session copy is retained and DOWNLOAD RECOVERY JSON is offered. Moving to a different address, browser or device requires JSON export/import.

Current English and German guides are linked directly in SETTINGS and the GitHub README. Downloaded PDFs are snapshots; the online links provide the current edition.

10 / SECTION

Set up AAR

Worked example: RHINE SENTINELLE / GHOST. Load optionally, never as a default.



1 / Position, time and flight condition

AAR after waypoint places the event after RP ANGIE. ARCT PLAN 17:50Z and AAR OUT PLAN 18:00Z are planned times. FL210, 280 KIAS and 3 BAGS determine the applicable consumption and transfer-rate model.

2 / Two independent choices

Transfer rate mode AUTO selects a rate from the available data. **Fixed / manual AAR onload** sets the fuel quantity instead. An AUTO rate and a manual 966 kg onload can be active together.

Keep estimates visible

AUTO REFERENCE ESTIMATE here means reference run S2, not a direct measurement at FL210 / 280 KIAS / 3 BAGS. The app calculates receiver fuel flow automatically. N/A / RATE REQUIRED means no usable rate and no verified transfer time.

11 / SECTION

Read the AAR ticket

Read fuel transfer, receiver burn and planned event duration separately.

AAR PLAN 01	
ARIP	NOT DEFINED
ARCP	RP ANGIE
ARCT	17:50Z
ALT	FL210
SPD	280 KIAS
CONF	3 BAGS
TANKER	A330 MRTT
CALLSIGN	MMF15
FREQ / STATUS	118.680 MHz TBC / provisional
ARCT / OUT PLAN	17:50Z / 18:00Z
FOB IN	6,256 KG
GROSS ONLOAD	+966 KG
TRANSFER RATE	488.02 KG/MIN EST
RATE SOURCE	AUTO REFERENCE ESTIMATE
EST XFER TIME	1.98 MIN
AAR EVENT	10.0 MIN
FOB AFTER LOAD	7,222 KG*
AAR BURN	-324 KG
FOB OUT	6,898 KG
TRANSFER / ORBIT BURN	64 / 260 KG
AAR OUT	18:00Z
BINGO 2	1,953 KG
B2 → RTB	ETNN
BINGO (NO AAR)	1,853 KG
JOKER	6,449 KG
JOKER WP	S-PUSH IN

FOB IN

6,256 kg before the event.

GROSS ONLOAD

966 kg from the tanker. This is not the net increase in tank contents.

TRANSFER RATE / EST XFER TIME

488.02 kg/min and 1.98 min are modeled estimates here. RATE SOURCE identifies the basis.

AAR EVENT

10 min overall: about 1.98 min transfer plus 8.02 min orbit/remaining time.

AAR BURN

324 kg total: 64 kg during transfer plus 260 kg during the remaining time.

FOB OUT / AAR OUT

6,898 kg at 18:00Z. The subsequent route continues from this result.

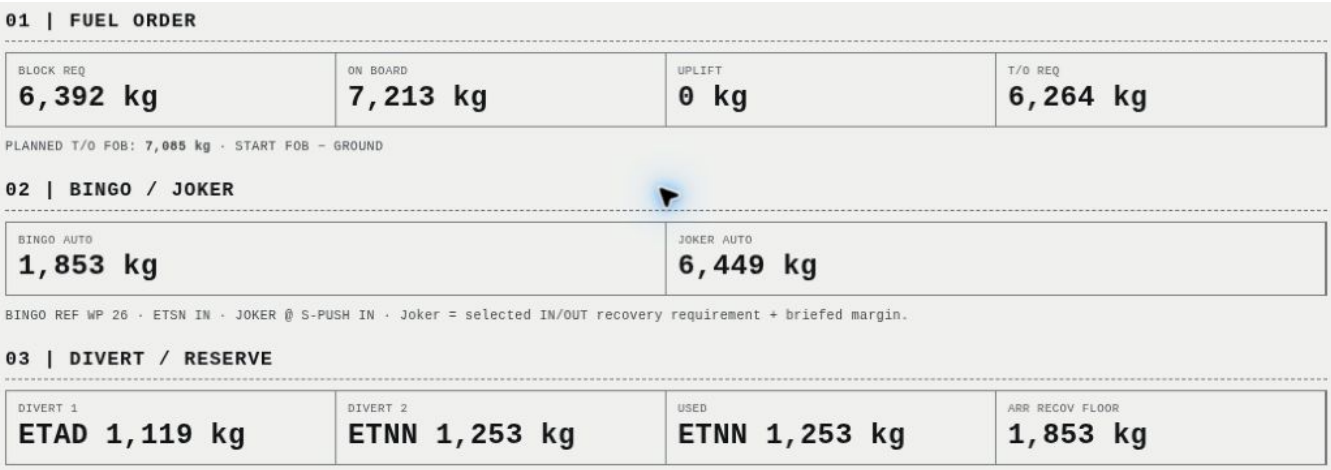
FOB AFTER LOAD = ACCOUNTING ONLY

7,222 kg is FOB IN + Gross Onload, not a physical tank state. Fuel is already being consumed during transfer. The capacity check uses actual tank contents. All example figures are rounded.

12 / SECTION

Fuel order, Joker and Bingo

Separate requirements, actual planned fuel and decision thresholds.



1 / REQUIRED is a minimum requirement

ON BOARD 7,213 kg is the entered start fuel. After ground fuel, PLANNED T/O FOB is 7,085 kg. T/O REQ 6,264 kg is the calculated take-off requirement. UPLIFT 0 kg means no additional ground uplift is required for this plan.

2 / Always read the reference point

JOKER 6,449 kg applies to **S-PUSH IN**. The hold there is still to be flown and is part of the remaining requirement. At WP OUT that hold would already be completed. The 795 kg Joker Buffer is an additional margin.

3 / USED selects the divert

USED ETNN: 1,253 kg plus 600 kg landing reserve gives a 1,853 kg recovery floor here. Bingo is referenced to arrival ETSN IN; remaining mission fuel there is zero. BINGO 2 at 1,953 kg and B2 → RTB ETNN are separately planned entries.

Not a universal threshold

Joker and Bingo change with route, reference point, hold, divert and reserves. These results belong to the loaded preset and current performance data.

13 / SECTION

Hold, WP IN and WP OUT

Flight log: from the AAR event through S-PUSH to the hard-time target.

05 | FLIGHT LOG

NR	WAYPOINT	ALT	M	WIND	LEG NM	ETE	WP IN Z	HOLD / WP OUT	LEG kg	RECOV kg	FOB IN / OUT	IST kg
01	ETSN	1,249	–	–	–	–	17:30Z PLAN 17:30Z	–	–	7,230	7,085	
02	PESIM	17,700	0.7	5	32.0	5.0	17:35Z	–	276	6,954	6,809	
03	MEXIT	21,000	0.7	10	52.1	7.3	17:42Z	–	273	6,681	6,536	
04	RP ANGIEAAR 966 kg ONLOAD	21,000	0.72	-7	56.3	7.7	17:50Z PLAN 17:50Z	–	280	6,401	IN 6,256 / OUT 6,898	
05	AAR EVENT	21,000	0.55	1	0.1	0.0	18:00Z	–	1	6,076	6,898	
06	AAR OUT	21,000	0.55	-10	0.1	0.0	18:00Z PLAN 18:00Z	–	1	6,076	6,897	
07	RUSTI / LL ENTRY	5,000	0.55	-12	23.1	4.8	18:05Z PLAN 18:05Z	–	103	5,973	6,794	
08	S-PUSH	2,000	0.55	-19	47.1	8.4	18:13Z PLAN 18:13Z	IN 18:13Z → OUT 18:30Z · 16.7m / 668kg	319	5,654	IN 6,475 / OUT 5,807	
09	LY	2,000	0.55	-15	28.1	4.9	18:35Z PLAN 18:35Z	–	194	4,791	5,612	
10	N11	2,000	0.55	-27	13.0	2.3	18:37Z PLAN 18:37Z	–	93	4,698	5,519	
11	LX	2,000	0.55	-26	24.4	4.4	18:42Z PLAN 18:42Z	–	174	4,524	5,345	
12	IP	2,000	0.58	-13	39.4	6.4	18:48Z PLAN 18:48Z	–	273	4,251	5,072	
13	TGT-SUIP	2,000	0.55	16	12.8	2.0	18:50Z HARD 18:50Z	–	81	4,170	4,991	

S-PUSH in detail

REFERENCE	TIME	MEANING
WP IN	18:13Z	FOB 6,475 kg before the hold.
HOLD	16.7 min	668 kg hold burn; display rounded.
WP OUT / GO	18:30Z	FOB 5,807 kg after the hold.
TGT-SUIP	18:50Z	HARD 18:50Z: controlling TOT anchor.

The hold shifts subsequent times and consumes fuel, but does not change coordinates or LEG NM. The 47.1 NM leg to S-PUSH remains unchanged. MIN EDIT / ETE describes the leg; HOLD is additional time at the waypoint.

Distinguish PLAN and HARD

PLAN marks a planned value. HARD identifies a binding time anchor. The preset already includes speed/hold adjustments for the TOT. M0.72 and M0.58 are rounded displays; full precision is retained internally. Investigate a new timing conflict rather than hiding it with altered distances.

14 / SECTION

Abbreviations & reading key

A quick reference for the terms used throughout the app.

TERM	MEANING IN THIS GUIDE
AAR	**Air-to-Air Refuelling** Air refuelling; simulator planning in this app.
ARIP	**Air Refuelling Initial Point** Initial point for the approach to refuelling.
ARCP	**Air Refuelling Control Point** Control/rendezvous reference point; RP ANGIE here.
ARCT	**Air Refuelling Control Time** Rendezvous time reference; the associated point depends on the procedure. See p. 15.
FOB	**Fuel on Board** Fuel carried; IN before, OUT after the event.
FL / KIAS	**Flight Level / Knots Indicated Airspeed** FL210 = 21,000 ft pressure altitude at standard pressure. KIAS is indicated airspeed.
CONF / 3 BAGS	**Configuration / three external tanks** Configuration / three external tanks. CLEAN = no external tanks.
EST / XFER	**Estimated / Transfer** Estimated / transfer. EST XFER TIME is estimated time with fuel flowing.
FREQ / TBC	**Frequency / To Be Confirmed** Radio frequency / to be confirmed; not a verified frequency assignment.
WP / ETA / ETE	**Waypoint / Estimated Time of Arrival / Estimated Time Enroute** Route point / arrival time / time enroute.
TOT / Z	**Time on Target / Zulu** Time at target / UTC time, not local daylight time.
B2 / RTB	**Bingo 2 / Return to Base** Separate second threshold / return to base.
REQ / T/O	**Required / Take-off** Required amount / take-off. Not the same as planned FOB.

Image source: official web app v1.9.23, 12 Oct 2026, optional RHINE SENTINELLE preset. Crops from actual screenshots, not recreated interfaces. This documentation update changes no app or mission values.

15 / SECTION

AAR: position, time, sequence

ARIP, ARCP and ARCT coordinate a rendezvous, not a fuel calculation.

Tanker and receiver have to meet in space and time. The three entries answer different questions. The terminology below comes from public US/civil aviation documents; the specific procedure is established in the applicable briefing. [1, 2]

ENTRY	PRACTICAL MEANING
ARIP	**Where does the rendezvous begin?** Air Refuelling Initial Point: an upstream geographical starting point that structures the entry towards the tanker rendezvous. [1]
ARCP	**Where should the formation come together?** Air Refuelling Control Point: planned location where the receiver reaches the observation/pre-contact position relative to the tanker. Fuel need not be flowing yet. [1]
ARCT	**When is that meeting planned?** Air Refuelling Control Time: the rendezvous time reference. It applies to the ARCP for a point-parallel rendezvous and to the ARIP for an enroute rendezvous. It is not universally the time at ARCP. [1]

Why it matters

Arrival in the refuelling area, rendezvous, stable contact, fuel flow and leaving the event are different stages. Real procedures also involve airspace coordination and clearances; reaching a time alone does not authorize refuelling. ARCT and time with fuel flowing are therefore not interchangeable. [2]

How the app represents it

AAR after waypoint locates the calculated event on the route. The ticket shows the calculated arrival there as ARCT. **ARCT PLAN** / **AAR OUT PLAN** are comparison entries, not substitutes for route hard times. ARIP/ARCP text fields create neither extra legs nor a rendezvous manoeuvre. [7]

SENTINELLE / RP ANGIE

ARIP remains NOT DEFINED because no point was supplied. RP ANGIE is the entered ARCP reference. 17:50Z to 18:00Z is the planned AAR block; this does not imply ten minutes of fuel transfer.

16 / SECTION

Gross, net and fuel on board

The engines keep running during refuelling. Fuel delivered and fuel gained are different quantities.

TERM	WHAT IT DESCRIBES
GROSS ONLOAD	Fuel transferred from tanker to receiver, in kg. Gross here refers to the delivery, not aircraft gross weight.
RECEIVER BURN	Fuel consumed by the receiving jet. Fuel flow is a rate in kg/min; burn is the quantity consumed over a period, in kg.
NET GAIN / FOB	Net gain is the increase between two comparable tank readings. FOB is the total fuel remaining at a specified moment.

```
FOB OUT = FOB IN + GROSS ONLOAD - EVENT BURN
NET GAIN = GROSS ONLOAD - EVENT BURN
```



Rounded Sentinel example: 6,256 + 966 - 324 = 6,898 kg. Although 966 kg is delivered, fuel increases by only 642 kg across the full event. Adding 966 kg to FOB IN without subtracting burn overstates onward fuel by 324 kg. [7]

Measurements: use matching start and end points

Tank readings and elapsed time first give net gain. To obtain gross quantity, add receiver burn over **that same interval**. If burn comes from a model, the derived gross rate remains an estimate. A pause without fuel flow must not silently be treated as continuous transfer time.

```
Gross rate = (FOB end - FOB start + interval burn) / minutes
```

“Gross fuel” is too ambiguous

Specify quantity, rate and reference time: “966 kg gross onload” is a delivery; “488 kg/min gross rate” is a flow rate; “FOB OUT 6,898 kg” is the result at event end.

17 / SECTION

Transfer time is not event time

Rate answers “how fast?”; event duration answers “how long overall?”.

An example from the ticket

PHASE	DURATION	FUEL EFFECT
Whole event	10.0 min	Consumption throughout the planned block.
Fuel transfer	1.98 min	About 1 min 59 s with fuel flowing; around 64 kg receiver burn.
Remaining time	8.02 min	Modeled as orbit/remaining time without delivery; around 260 kg burn.

The example effective gross rate of 488.02 kg/min is much higher than the modeled receiver flow of 32.41 kg/min. During transfer, FOB therefore rises at about 455.61 kg/min net. Once transfer stops, FOB falls again during the remaining time. Displayed values are rounded. [7]

Why TARGET FOB OUT belongs at event end

A requested target must remain after the whole event. Waiting after contact consumes additional fuel. With constant transfer rate R, receiver flow F and planned event duration D:

```
G = max(0, Target OUT - FOB IN + F x D)
```

```
Transfer time = G / R (must fit inside D)
```

Only when there is no additional time does a positive required gain give t = (Target - FOB IN) / (R - F). A measured rate profile that varies with fuel state is evaluated by segments instead. This explanation changes no app formula. [7]

Check capacity at the actual peak

FOB AFTER LOAD 7,222 kg is simply IN + delivery. Transfer burn has already occurred: about 7,158 kg at transfer end, then about 6,898 kg at event end. The modeled actual peak stays below the 7,213 kg capacity. Waiting after transfer would not retroactively fix a genuine earlier overflow. [7]

Keep model limits visible

N/A / RATE REQUIRED: no supported transfer time. RATE ≤ RECEIVER FLOW: no positive net gain in this model. EVENT TOO SHORT: insufficient event time. AUTO REFERENCE ESTIMATE: reference assumption, not a directly measured condition.

18 / SECTION

Understand Bingo and Joker

Fuel thresholds are decision points with a reference, not properties of a tank.

TERM	MEANING
BINGO	In US multi-service terminology, the briefed fuel state required for recovery. [3]
JOKER	A higher state at which separation, withdrawal or event termination should begin. The briefed follow-on action matters. [3]

The number depends on the plan. A public USAF training manual describes multiple Joker values at transition points and adjustments for changed flight conditions. This illustrates the briefing principle; it does not transfer T-6 fuel figures to a Eurofighter. [4]

What “recovery” means here

Return through the intended landing, with the planned reserves and alternate where applicable. Bingo therefore does not mean “almost empty” or landing reserve alone. Recovery begins at Bingo so that the required reserve remains at landing. [4]

The specific implementation in this app

At the selected reference WP, the app accounts for **remaining planned mission + Used Divert + Recovery Extra + Landing Reserve**. This is not a universal military direct-RTB formula. If the intended action is to fly directly home, the assumed route must match that action. [7]

JOKER @ WP IN	The hold at the reference point is still ahead; its burn is included.
JOKER @ WP OUT	That hold is complete; its burn must not be added again.
JOKER BUFFER	An additional planning margin, not a universal fixed value or a substitute for divert/reserve.

Sentinel: interpret a small margin correctly

S-PUSH IN FOB 6,475 kg versus Joker IN 6,449 kg: only about 26 kg above the planned threshold. That is not 26 kg until an empty tank. Bingo 1,853 kg here is referenced to arrival ETSN IN and must not be read as the same requirement everywhere along the route. [7]

19 / SECTION

Tiger and the AAR perspective

What the terms say about readiness, and what the app does not infer from them.

TIGER / TIGER FUEL

TIGER reports sufficient **fuel and ordnance** to accept a commit: an assigned engagement/task. The definition gives no fixed fuel quantity. [3]

It follows that a full tank alone does not establish this status. A “Tiger fuel” figure needs an agreed simulator briefing meaning.

In this app: background knowledge

The app calculates fuel but does not assess ordnance or ability to accept a commit. It therefore provides no automatically calculated TIGER status. BINGO 2 is not hidden Tiger fuel either. This guide adds explanation only. [7]

Real-world context: Typhoon and A330

The RAF describes Typhoon as using probe-and-drogue refuelling. Its A330-based Voyager uses hose systems, with underwing stations for fast jets. A tanker name alone therefore does not specify an effective receiver rate. This app uses simulator measurements and identified models, not real-world performance approval. [5, 6, 7]

What to agree before a shared simulator flight

Meeting point and time	Which point is meant? Which time is PLAN and which HARD? Is an ARIP defined at all?
Fuel request	Gross quantity or target FOB at event end? kg or lb? How much time is allocated to the entire event?
Recovery decision	Which WP does Joker/Bingo apply to? IN or OUT? Which divert is USED and which reserve is planned?
If AAR is unavailable	Check a separate simulator case without onload. Planned transfer is not fuel already delivered.

20 / SECTION

Sources and scope

Public background, worked examples and application behaviour are identified separately.

[1] USAF - AFMAN 11-255, Volume 3

16 Nov 2023, Attachment 1, pp. 42-43
ARIP, ARCP and procedure-dependent ARCT definitions.
Open primary source

[2] FAA - Special Operations, Chapter 5

Section 1, General / Chronology
Rendezvous, coordination and distinction from actual transfer.
Open primary source

[3] ALSSA - Multi-Service Brevity Codes

25 Apr 2025, Table 2: BINGO p. 5, JOKER p. 25, TIGER p. 50
US multi-service meanings; not assumed identical to every NATO or national procedure.
Open primary source

[4] USAF - AETCMAN 11-248

13 Aug 2025, section 1.11, Fuel Considerations
Briefing and variable Joker/Bingo references, illustrated by a training aircraft.
Open primary source

[5] Royal Air Force - Typhoon FGR4

Aircraft overview / Air-to-Air Refuelling
Probe-and-drogue system.
Open primary source

[6] Royal Air Force - Voyager

Aircraft overview / Role
A330 basis and hose stations for different receivers.
Open primary source

[7] Eurofighter IFE Fuel Calculator v1.9.23

Application code: app.js, logic-core.js, AAR profile model and optional RHINE SENTINELLE preset. Screenshots and worked values from the official web app. Equations and explanations are original descriptions of the implemented model.

Scope

Checked 12 Oct 2026. These background chapters explain concepts for MSFS / IndiaFoxtEcho. National procedures and specific briefings may differ. This is not real-world Eurofighter flight, refuelling or mission authorization. App logic, mission values and defaults remain unchanged.